



TOUFIK MAHATA

B.Tech Biotechnology Student | Aspiring Bioinformatics Scientist

Samastipur, Bihar, India | toufikmahata20@gmail.com | [+91 6296156961]

[LinkedIn](#) | [ResearchGate](#) | [GitHub](#) | [GeeksforGeeks](#)

PROFILE

B.Tech Biotechnology student at Dr. Rajendra Prasad Central Agricultural University with a strong interest in the intersection of biology, data, and computation. Author of multiple research articles spanning population genetics, agri-tech AI systems, and origin-of-life chemistry, alongside hands-on software development experience in Python, JavaScript, and embedded systems. Working toward a research career in bioinformatics, genomics, and computational biology.

EDUCATION

B.Tech in Biotechnology

Dr. Rajendra Prasad Central Agricultural University, Samastipur, Bihar

LEADERSHIP & EXTRACURRICULAR

Campus Mantri — GeeksforGeeks

- Selected as campus ambassador for GeeksforGeeks, representing the platform at Dr. Rajendra Prasad Central Agricultural University.
- Lead awareness drives, workshops, and coding-culture initiatives on campus, and serve as a bridge between GeeksforGeeks and the student community.

RESEARCH PUBLICATIONS

- Population Genetics and Genetic Drift** — co-authored with Ayush Gupta and Anjana Kumari; explores theoretical and simulation-based models of genetic drift, the Wright-Fisher model, fixation probability, and the bottleneck effect. (Mar 2026)
- Soil Microbiome AI: A Mobile-Based Intelligent System for Predicting Soil Biological Health Without DNA Sequencing** — sole author; proposes a mobile-supported alternative to costly DNA-based metagenomic sequencing for assessing agricultural soil health. (Dec 2025)
- The Ribose Paradox: A Systems-Chemistry Critique of the Naked-Replicator Model** — sole author; a critical systems-chemistry analysis of the RNA World hypothesis and the thermodynamic challenges of prebiotic ribose stability. (Dec 2025)
- AI Crop Moisture Stress & Irrigation Advisory System** — sole author; research proposal on an AI-driven methodology for crop-type identification, moisture-stress detection, and irrigation advisory using spectral and SAR data. (Jun 2026)

Full publication list and citations available on [ResearchGate](#).

PROJECTS

- Soil Microbiome AI** — mobile-based intelligent system for predicting soil biological health as a low-cost alternative to DNA sequencing; published as a full research article with an accompanying app.
 - Bluetooth-Controlled Obstacle-Avoiding RC Car** — ESP32-based remote-control car with real-time obstacle detection and avoidance (C++).
 - Student Essentials Delivery App** — concept and build for a delivery platform addressing everyday student needs on campus.
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SKILLS

Biotechnology & Life Sciences: Molecular Biotechnology, Genetic Engineering, Molecular Genetics, rDNA Technology, DNA Amplification, Biomedical Science, Cancer Biology, Population Genetics

Computational & Bioinformatics: Data-driven biological analysis, genomics-oriented computation, AI/ML applications in agriculture and life sciences

Programming & Tools: Python, JavaScript, HTML/CSS, C/C++, Git, Node.js, problem solving & DSA practice (GeeksforGeeks)

RESEARCH INTERESTS

Genomics · Molecular Biology · Computational Biology · Bioinformatics · Population Genetics · Agricultural Biotechnology & AgriTech AI

CERTIFICATIONS & TRAINING

- **Google Analytics Certification** — Google Digital Academy (Skillshop) (Jan 2026)
 - **Certificate Course in Genetics** — Reliance Foundation Skilling Academy (Jan 2026)
 - **Generative AI Mastermind Program** — Outskill (Jun 2026)
 - **Drowsiness Detection using OpenCV** — NoviTech R&D Pvt Ltd (Feb 2026)
 - **Gemini Certification for Students (K12)** — United Latino Students Association (Jan 2026)
 - **Investment Banking Job Simulation** — J.P. Morgan Chase, via Forage (Jan 2026)
 - **Software Engineering Job Simulation** — J.P. Morgan Chase, via Forage (Jan 2026)
- **Quantitative Research Job Simulation** — J.P. Morgan, via Forage (Jan 2026)
 - **Concept to Creation — Jersey Design** — Unstop (Royals HunaRR Manch S2) (May 2026)
 - **Certificate of Midnight Explorer** — HackQuest (Jan 2026)
 - **Bridge4Change Foundation Program** — Bridge4Change Foundation (Dec 2025)
 - **All India Career Summit — Participation** — Aptus Technologies (Feb 2026)
 - **Career Guidance Session — Participation** — GradGuru Innovations (Feb 2026)
 - **Quiz on Policing and Law Enforcement** — MyGov India